

HYDRA Documentation

Installation Guide Oracle 19c for SLES12 for HYDRA MW4.0pe

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1 Introduction

Attention:

To perform the following procedures advanced knowledge of your server operating system, the intended database software and common IT systems is required.

These procedures are performed on your own responsibility.

MPDV Mikrolab GmbH is not liable for any loss of or destruction of data.

In case of doubt place an order with MPDV Mikrolab GmbH to perform these tasks for you.

This manual explains how to install an Oracle Database 19c (19.3.0.0.0) on a server with a SUSE Linux Enterprise Server 12 SP5 (SLES12 SP5) operating system for use with MPDVs Manufacturing Integration Platform (**MIP**) based products like MIP 1.1, HYDRA MW4.0pe or FEDRA 1.1.

Other than MIP the applications HYDRA and FEDRA allow for a multi system installation (multiple HYDRA or FEDRA systems on the same (application) server).

For multi system installations every HYDRA or FEDRA system needs its own database.

Name your databases **mip1**, **mip2**, **mip3**, etc.

With multi system installations it is strongly recommended that every database has its own database instance available, e.g.: **MIP1**, **MIP2**, **MIP3**, etc.

If the database installation is meant as preliminary installation for MPDV personnel to install a MIP based product on your server you can stop as soon as you are finished with chapter "[5 Installing Oracle 32Bit Client](#)".

Attention: Do not run these installation procedures on a computer where MIP based products like MIP 1.1, HYDRA MW4.0pe or FEDRA 1.1 are already running.

By doing so, you might destroy your existing MIP based system.

Please be extremely cautious if you decide to do it anyway.

2 Installation Requirements

- Server hardware and software according to the recommendations of MPDV Mikrolab GmbH for the respective MIP based product.

See manuals: **HW_SW_GUIDE.pdf**

- English operating system:

SUSE Linux Enterprise Server 12 SP5 (x86_64)

- **Important:** Server configuration according to MPDVs server configuration manuals for the respective MIP based product, e.g.:

"Preparation Guide for SLES12 SP5" (e.g.: PreInstGuide_MW40_SLES12_SP5.pdf)

- Installation files for the required edition of **"Oracle Database 19c (19.3) for Linux x86-64"**

LINUX.X64_193000_db_home.zip

Link: <https://www.oracle.com/database/technologies/oracle19c-linux-downloads.html>

Oracle Database 19c (19.3)

Oracle Database 19c (19.3) for Linux x86-64 (RPM)

Download	Description
 oracle-database-ee-19c-1.0-1.x86_64.rpm	(2,694,664,264 bytes) (sha256sum - c519397aea7af9f097cbde6597783f17964e5d53ea3fd90da042d4b65d379652)

- Installation files for the client software " Oracle Database 19c Client (19.3) for Linux x86-64"

LINUX.X64_193000_client_home.zip

Link: <http://www.oracle.com/technetwork/database/enterprise-edition/downloads/database19c-linux-download-2240591.html>

Oracle Database 19c Client (19.3) for Linux x86-64

Download	Description
 LINUX.X64_193000_client.zip	(64-bit) (1,134,912,540 bytes) (sha256sum - ee33637947d760413790d16bd11fc273a6e8c0ee3193e215a85af4c9dd1a5834)
 LINUX_193000_client.zip	(32-bit) (1,018,382,779 bytes) (sha256sum - 4ec2b1d4e311febddcbb3597050f20c5c7afe30e061fbd84e42060b7582c59a)
 LINUX.X64_193000_client_home.zip	(64-bit) (987,216,967 bytes) (sha256sum - df3ef12bc19d03cb331ccdf3490ec361b6ed5d243da32df583f02f0774e2315)
 LINUX_193000_client_home.zip	(32-bit) (917,311,326 bytes) (sha256sum - 256f01d782e884dce3bf0f515c8b562b22105c1698cf6a9e6cf7acc47e7f959e)

- Minimum hard disk configuration:
Partition \ (**root**) with a minimum of **100 GB** disk space available and file system **EXT3** for the LINUX installation
Partition **lu1** with a minimum of **200 GB** disk space available and file system **EXT3** for the installation of the MIP based product
Partition **lu2** with a minimum of **400 GB** disk space available and file system **EXT3** for database data (default configuration)
- Note: If you are using a different disk layout you must make sure that it matches the configuration used in the database installation scripts (see chapter [6.1 Create MIP Database](#)).
- Screen resolution with min. **1024x768** pixel.
- Local LINUX users **mipadm** and **oracle** must be available.
(see server configuration manual, e.g. PreInstGuide_MW40_SLES12_SP5.pdf)

3 Installing Oracle 19c Database

3.1 Preinstallation Tasks

Login as user "root".

Check the file `/etc/hosts`

If there is the hostname with the IP-address 127.0.0.2 in `/etc/hosts` deactivate or delete that line, e.g.:

```
# 127.0.0.2          linux01.mpdv.local linux01
```

Make sure that `/etc/hosts` contains the hostname of your server with its correct IP-Address, e.g.:

```
192.168.20.232      linux01.mpdv.local linux01
```

Edit the following files belonging to the software package **orarun**:

`/etc/sysconfig/oracle`

```
ORACLE_BASE=/u1/oracle
```

```
START_ORACLE_DB="yes"
```

```
START_ORACLE_DB_LISTENER="yes"
```

```
START_ORACLE_DB_EMANAGER="yes"
```

```
SHMMAX= Recommended: SHMMAX = 0.5*(physical memory)
```

```
example for 24GB RAM: SHMMAX=12884901888
```

```
do not use values smaller than SHMMAX=4294967295
```

`/etc/profile.d/oracle.sh`

```
ORACLE_BASE=/u1/oracle
```

```
ORACLE_HOME=$ORACLE_BASE/ora19
```

```
ORACLE_SID=MIP1
```

Check that the following settings are disabled:

```
# export ORA_CRS_HOME=$ORACLE_BASE/product/12cR1/crs
```

```
# export ORA_ASM_HOME=$ORACLE_BASE/product/12cR1/asm
```

`/etc/profile.d/oracle.csh`

```
setenv ORACLE_BASE /u1/oracle
```

```
setenv ORACLE_HOME ${ORACLE_BASE}/ora19
```

```
setenv ORACLE_SID MIP1
```

Check that the following settings are disabled:

```
# setenv ORA_CRS_HOME ${ORACLE_BASE}/product/12cR1/crs
```

```
# setenv ORA_ASM_HOME ${ORACLE_BASE}/product/12cR1/asm
```

Activate the new kernel parameters by running the following command as user **root**:

```
cd /etc/init.d
./oracle start
```

Check the actual kernel parameter with the following commands:

```
/sbin/sysctl -a | grep aio-max-nr
/sbin/sysctl -a | grep file-max
/sbin/sysctl -a | grep shm
/sbin/sysctl -a | grep sem
/sbin/sysctl -a | grep ip_local_port_range
/sbin/sysctl -a | grep rmem_default
/sbin/sysctl -a | grep rmem_max
/sbin/sysctl -a | grep wmem_default
/sbin/sysctl -a | grep wmem_max
/sbin/sysctl -a | grep kernel.panic_on_oops
```

The following values must be set at least to:

```
fs.aio-max-nr = 1048576
fs.file-max = 6815744
kernel.shmmax = minimum 4294967295 or more
kernel.shmall = 2097152
kernel.shmmni = 4096
kernel.sem = 1250 32000 100 256
net.ipv4.ip_local_port_range = 9000 65535
net.core.rmem_default = 262144
net.core.rmem_max = 4194304
net.core.wmem_default = 262144
net.core.wmem_max = 1048576
kernel.panic_on_oops = 1
```

If the actual values are too small or missing completely then they must be set in the file:

```
/etc/sysctl.conf
```

Edit the file and add the desired parameter(s), e.g.:


```
kernel.shmmax = 4294967295
net.ipv4.ip_local_port_range = 9000 65535
kernel.panic_on_oops = 1
```

After editing the file it has to be ensured that the SUSE Enterprise Server will read the file /etc/sysctl.conf and that the changes are available in the active kernel memory:

```
/sbin/sysctl -p
```

With the following command you can check if the values are set correctly:

```
/sbin/sysctl -a
```

Login as user "**oracle**" at the KDE desktop.

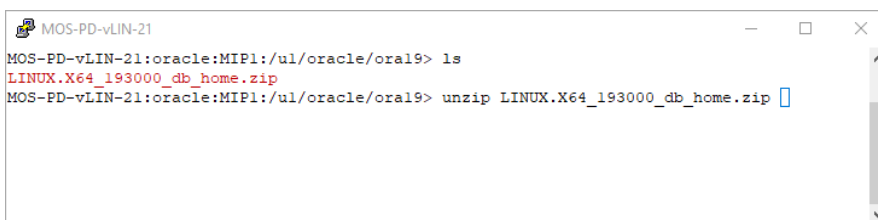
Do not "su – oracle" from another user account!

Copy the installation file **LINUX.X64_193000_db_home.zip** to the directory \$ORACLE_HOME on the server, e.g. to: **/u1/oracle/ora19**

```
cp /u1/install/LINUX.X64_193000_db_home.zip /u1/oracle/ora19
```

Unzip the installation file to your intended ORACLE_HOME location, e.g. **/u1/oracle/ora19**:

```
cd /u1/oracle/ora19
unzip LINUX.X64_193000_db_home.zip
```



Change the following settings inside /u1/oracle/ora19/cv/cvdata/cvu_prereq.xml

Delete the following lines:

```
<PACKAGE NAME="gcc-c++" VALUE="4.8" SEVERITY="IGNORABLE" ARCHITECTURE="x86_64"/>
<PACKAGE NAME="libstdc++33" VALUE="3.3.3-62.1" SEVERITY="CRITICAL" ARCHITECTURE="x86_64"/>
<PACKAGE NAME="gcc-c++" VALUE="7" SEVERITY="IGNORABLE" ARCHITECTURE="x86_64"/>
```

```
<PACKAGE NAME="libstdc++33" VALUE="3.3.3-62.1" SEVERITY="CRITICAL"/>
<PACKAGE NAME="gcc-c++-32bit" VALUE="7-1.563" SEVERITY="IGNORABLE"/>
<PACKAGE NAME="gcc-c++" VALUE="7-1.563" SEVERITY="IGNORABLE"/>
<PACKAGE NAME="gcc-32bit" VALUE="7-1.563" SEVERITY="IGNORABLE"/>
<PACKAGE NAME="gcc" VALUE="7-1.563" SEVERITY="IGNORABLE"/>
<PACKAGE NAME="JDK" VALUE="1.8.0.5.151" SEVERITY="IGNORABLE"/>
```

Edit the following lines:

```
<PACKAGE NAME="libstdc++6" VALUE="4.8.3" SEVERITY="IGNORABLE"/> to "CRITICAL"

<PACKAGE NAME="libstdc++6" VALUE="7.3.1" SEVERITY="IGNORABLE"/> to "CRITICAL"

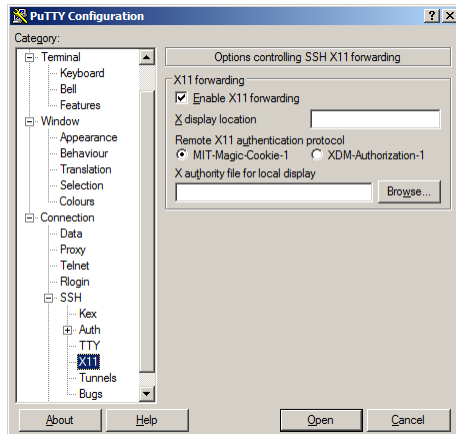
<PACKAGE NAME="libpcre16" VALUE="0-8.41" SEVERITY="IGNORABLE"/> to
<PACKAGE NAME="libpcre16-0" VALUE="8.41" SEVERITY="IGNORABLE"/>
```

Note: Alternatively the installation can be executed remotely.

Therefore a X-Windows program must run on the executing desktop PC (e.g. Xming).

After that a connection via `putty` must be initiated to the appropriate system.

In the "Session settings" the checkbox "**Enable X11 forwarding**" must be set.



For testing purpose you can start the program "xclock".

You should see the following window appearing at your PC.

Now you can start the installation as described below.



3.2 Database Software Installation

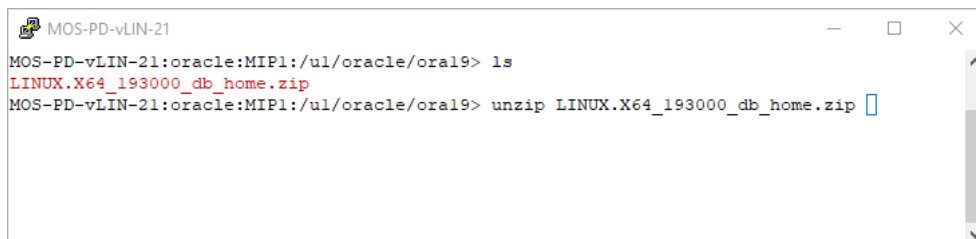
Before starting the installation of the Oracle database software it is recommended to review the standard installation documentation from Oracle. Especially regarding the topic installation of multiple Oracle database and client versions of the same hardware, e.g.:

<https://docs.oracle.com/en/database/oracle/oracle-database/19/ladbi/database-installation-guide-linux.pdf>

<https://docs.oracle.com/en/database/oracle/oracle-database/19/lacli/database-client-installation-guide-linux.pdf>

This document only covers the installation of the Oracle 19c database software on a system without any other installed versions.

Unzip the Oracle installation file to your intended ORACLE_HOME location ({hostname}:oracle:MIP1:**/u1/oracle/ora19**):



```

MOS-PD-vLIN-21
MOS-PD-vLIN-21:oracle:MIP1:/u1/oracle/ora19> ls
LINUX.X64_193000_db_home.zip
MOS-PD-vLIN-21:oracle:MIP1:/u1/oracle/ora19> unzip LINUX.X64_193000_db_home.zip

```

After the extraction of the Oracle ZIP file you have to adapt the Oracle database response file before you can proceed with the installation.

The installation will be executed as a silent installation.

The response file is stored at:

```
/u1/mip1/db_sql/oracle19/linux/db_inst/ora19db_inst.rsp
```

Before you could proceed with the installation some (red) values must be checked.

The preconfigured values comply to the MPDV recommendations.

Dissenting installations could be performed in charge of the customer.

```

#####
## Copyright(c) Oracle Corporation 1998,2019. All rights reserved.      ##
##                                                                    ##
## Specify values for the variables listed below to customize          ##
## your installation.                                                    ##
##                                                                    ##
## Each variable is associated with a comment. The comment             ##
## can help to populate the variables with the appropriate              ##
## values.                                                                ##
##                                                                    ##
## IMPORTANT NOTE: This file contains plain text passwords and         ##
## should be secured to have read permission only by oracle user       ##

```

```

## or db administrator who owns this installation.                ##
##                                                                ##
#####

#-----
# Do not change the following system generated value.
#-----
oracle.install.responseFileVersion=/oracle/install/rspfmt_dbinstall_response_schema_v19.0.0

#-----
# Specify the installation option.
# It can be one of the following:
# - INSTALL_DB_SWONLY
# - INSTALL_DB_AND_CONFIG
#-----
oracle.install.option=INSTALL_DB_SWONLY

#-----
# Specify the Unix group to be set for the inventory directory.
#-----
UNIX_GROUP_NAME=oinstall

#-----
# Specify the location which holds the inventory files.
# This is an optional parameter if installing on
# Windows based Operating System.
#-----
INVENTORY_LOCATION=/u1/oraInventory

#-----
# Specify the complete path of the Oracle Base.
#-----
ORACLE_BASE=/u1/oracle

#-----
# Specify the installation edition of the component.
#
# The value should contain only one of these choices.

# - EE   : Enterprise Edition

# - SE2   : Standard Edition 2

#-----

oracle.install.db.InstallEdition=SE2
#####
#
# PRIVILEGED OPERATING SYSTEM GROUPS
#-----
# Provide values for the OS groups to which SYSDBA and SYSOPER privileges
# needs to be granted. If the install is being performed as a member of the
# group "dba", then that will be used unless specified otherwise below.
#
# The value to be specified for OSDBA and OSOPER group is only for UNIX based
# Operating System.
#
#####
#-----
# The OSDBA_GROUP is the OS group which is to be granted SYSDBA privileges.
#-----
oracle.install.db.OSDBA_GROUP=dba
#-----

```

```

# The OSOPER_GROUP is the OS group which is to be granted SYSOPER privileges.
# The value to be specified for OSOPER group is optional.
#-----
oracle.install.db.OSOPER_GROUP=dba

#-----
# The OSBACKUPDBA_GROUP is the OS group which is to be granted SYSBACKUP privileges.
#-----
oracle.install.db.OSBACKUPDBA_GROUP=dba

#-----
# The OSDGDBA_GROUP is the OS group which is to be granted SYSDBG privileges.
#-----
oracle.install.db.OSDGDBA_GROUP=dba

#-----
# The OSKMDBA_GROUP is the OS group which is to be granted SYSKM privileges.
#-----
oracle.install.db.OSKMDBA_GROUP=dba

#-----
# The OSRACDBA_GROUP is the OS group which is to be granted SYSRAC privileges.
#-----
oracle.install.db.OSRACDBA_GROUP=dba
#####
#                                     #
#           Root script execution configuration           #
#                                     #
#####

#-----
# Specify the root script execution mode.
#
# - true : To execute the root script automatically by using the appropriate configuration methods.
# - false : To execute the root script manually.
#
# If this option is selected, password should be specified on the console.
#-----
oracle.install.db.rootconfig.executeRootScript=true

#-----
# Specify the configuration method to be used for automatic root script execution.
#
# Following are the possible choices:
# - ROOT
# - SUDO
#-----
oracle.install.db.rootconfig.configMethod=ROOT
#-----
# Specify the absolute path of the sudo program.
#
# Applicable only when SUDO configuration method was chosen.
#-----
oracle.install.db.rootconfig.sudoPath=

#-----
# Specify the name of the user who is in the sudoers list.
# Applicable only when SUDO configuration method was chosen.
# Note:For Single Instance database installations,the sudo user name must be the username of the user installing the
database.
#-----
oracle.install.db.rootconfig.sudoUserName=

#####
#                                     #
#           Grid Options           #
#                                     #
#

```

```
#####

#-----
# Value is required only if the specified install option is INSTALL_DB_SWONLY
#
# Specify the cluster node names selected during the installation.
#
# Example : oracle.install.db.CLUSTER_NODES=node1,node2
#-----
oracle.install.db.CLUSTER_NODES=

#####
#                                     #
#       Database Configuration Options       #
#                                     #
#####

#-----
# Specify the type of database to create.
# It can be one of the following:
# - GENERAL_PURPOSE
# - DATA_WAREHOUSE
# GENERAL_PURPOSE: A starter database designed for general purpose use or transaction-heavy applications.
# DATA_WAREHOUSE : A starter database optimized for data warehousing applications.
#-----
oracle.install.db.config.starterdb.type=GENERAL_PURPOSE

#-----
# Specify the Starter Database Global Database Name.
#-----
oracle.install.db.config.starterdb.globalDBName=

#-----
# Specify the Starter Database SID.
#-----
oracle.install.db.config.starterdb.SID=

#-----
# Specify whether the database should be configured as a Container database.
# The value can be either "true" or "false". If left blank it will be assumed
# to be "false".
#-----
oracle.install.db.ConfigureAsContainerDB=false

#-----
# Specify the Pluggable Database name for the pluggable database in Container Database.
#-----
oracle.install.db.config.PDBName=

#-----
# Specify the Starter Database character set.
#
# One of the following
# AL32UTF8, WE8ISO8859P15, WE8MSWIN1252, EE8ISO8859P2,
# EE8MSWIN1250, NE8ISO8859P10, NEE8ISO8859P4, BLT8MSWIN1257,
# BLT8ISO8859P13, CL8ISO8859P5, CL8MSWIN1251, AR8ISO8859P6,
# AR8MSWIN1256, EL8ISO8859P7, EL8MSWIN1253, IW8ISO8859P8,
# IW8MSWIN1255, JA16EUC, JA16EUCTILDE, JA16SJIS, JA16SJISTILDE,
# KO16MSWIN949, ZHS16GBK, TH8TISASCII, ZHT32EUC, ZHT16MSWIN950,
# ZHT16HKSCS, WE8ISO8859P9, TR8MSWIN1254, VN8MSWIN1258
#-----
oracle.install.db.config.starterdb.characterSet=

#-----
# This variable should be set to true if Automatic Memory Management
# in Database is desired.
# If Automatic Memory Management is not desired, and memory allocation
```

```

# is to be done manually, then set it to false.
#-----
oracle.install.db.config.starterdb.memoryOption=false

#-----
# Specify the total memory allocation for the database. Value(in MB) should be
# at least 256 MB, and should not exceed the total physical memory available
# on the system.
# Example: oracle.install.db.config.starterdb.memoryLimit=512
#-----
oracle.install.db.config.starterdb.memoryLimit=

#-----
# This variable controls whether to load Example Schemas onto
# the starter database or not.
# The value can be either "true" or "false". If left blank it will be assumed
# to be "false".
#-----
oracle.install.db.config.starterdb.installExampleSchemas=false

#####
#
# Passwords can be supplied for the following four schemas in the
# starter database:
# SYS
# SYSTEM
# DBSNMP (used by Enterprise Manager)
#
# Same password can be used for all accounts (not recommended)
# or different passwords for each account can be provided (recommended)
#
#####

#-----
# This variable holds the password that is to be used for all schemas in the
# starter database.
#-----
oracle.install.db.config.starterdb.password.ALL=

#-----
# Specify the SYS password for the starter database.
#-----
oracle.install.db.config.starterdb.password.SYS=

#-----
# Specify the SYSTEM password for the starter database.
#-----
oracle.install.db.config.starterdb.password.SYSTEM=

#-----
# Specify the DBSNMP password for the starter database.
# Applicable only when oracle.install.db.config.starterdb.managementOption=CLOUD_CONTROL
#-----
oracle.install.db.config.starterdb.password.DBSNMP=

#-----
# Specify the PDBADMIN password required for creation of Pluggable Database in the Container Database.
#-----
oracle.install.db.config.starterdb.password.PDBADMIN=

#-----
# Specify the management option to use for managing the database.
# Options are:
# 1. CLOUD_CONTROL - If you want to manage your database with Enterprise Manager Cloud Control along with Database
# Express.
# 2. DEFAULT -If you want to manage your database using the default Database Express option.
#-----

```



```

oracle.install.db.config.starterdb.managementOption=DEFAULT

#-----
# Specify the OMS host to connect to Cloud Control.
# Applicable only when oracle.install.db.config.starterdb.managementOption=CLOUD_CONTROL
#-----
oracle.install.db.config.starterdb.omsHost=

#-----
# Specify the OMS port to connect to Cloud Control.
# Applicable only when oracle.install.db.config.starterdb.managementOption=CLOUD_CONTROL
#-----
oracle.install.db.config.starterdb.omsPort=0

#-----
# Specify the EM Admin user name to use to connect to Cloud Control.
# Applicable only when oracle.install.db.config.starterdb.managementOption=CLOUD_CONTROL
#-----
oracle.install.db.config.starterdb.emAdminUser=

#-----
# Specify the EM Admin password to use to connect to Cloud Control.
# Applicable only when oracle.install.db.config.starterdb.managementOption=CLOUD_CONTROL
#-----
oracle.install.db.config.starterdb.emAdminPassword=

#####
#                                                                    #
# SPECIFY RECOVERY OPTIONS                                          #
# -----                                                         #
# Recovery options for the database can be mentioned using the entries below
#                                                                    #
#####

#-----
# This variable is to be set to false if database recovery is not required. Else
# this can be set to true.
#-----
oracle.install.db.config.starterdb.enableRecovery=false

#-----
# Specify the type of storage to use for the database.
# It can be one of the following:
# - FILE_SYSTEM_STORAGE
# - ASM_STORAGE
#-----
oracle.install.db.config.starterdb.storageType=

#-----
# Specify the database file location which is a directory for datafiles, control
# files, redo logs.
#
# Applicable only when oracle.install.db.config.starterdb.storage=FILE_SYSTEM_STORAGE
#-----
oracle.install.db.config.starterdb.fileSystemStorage.dataLocation=

#-----
# Specify the recovery location.
#
# Applicable only when oracle.install.db.config.starterdb.storage=FILE_SYSTEM_STORAGE
#-----
oracle.install.db.config.starterdb.fileSystemStorage.recoveryLocation=

#-----
# Specify the existing ASM disk groups to be used for storage.
#
# Applicable only when oracle.install.db.config.starterdb.storageType=ASM_STORAGE

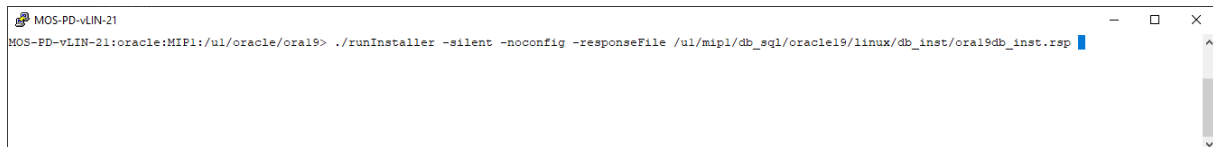
```

```
#-----  
oracle.install.db.config.asm.diskGroup=  
  
#-----  
# Specify the password for ASMSNMP user of the ASM instance.  
#  
# Applicable only when oracle.install.db.config.starterdb.storage=ASM_STORAGE  
#-----  
oracle.install.db.config.asm.ASMNMPPassword=
```

Change to the ORACLE_HOME directory (e.g. `cd /u1/oracle/ora19`)

Start the installation:

```
./runInstaller -silent -noconfig -responseFile  
/u1/mip1/db_sql/oracle19/linux/db_inst/ora19db_inst.rsp
```



A terminal window titled 'MOS-PD-vLIN-21' showing the command `./runInstaller -silent -noconfig -responseFile /u1/mip1/db_sql/oracle19/linux/db_inst/ora19db_inst.rsp` being executed. The cursor is at the end of the command line.

During the installation you will be asked for the “root” password



A terminal window titled 'MOS-PD-vLIN-21' showing the output of the Oracle Database Setup Wizard. The output includes the command `cd ora19/`, the file `ra19db_inst.rsp`, and the message 'Launching Oracle Database Setup Wizard...'. A red box highlights the prompt 'Enter password for 'root' user:'. Below this, the response file path is shown: `/u1/oracle/ora19/install/response/db_2020-03-02_03-05-01PM.rsp`. The log path is also shown: `/tmp/InstallActions2020-03-02_03-05-01PM/installActions2020-03-02_03-05-01PM.log`. A green box highlights the message 'Successfully Setup Software.'. The final message is 'Moved the install session logs to: /u1/oraInventory/logs/InstallActions2020-03-02_03-05-01PM'.

4 Installing Oracle Patch Set

The latest Oracle patch set approved by MPDV Mikrolab GmbH must be installed.

Version: **currently there is no approved patch set to be installed**

Please take note of the "Patch Set Notes" delivered with the patch set.

5 Post Installation Configuration

To perform the following steps either the MIP based application software must already be available on the server or the following files must be provided to you by MPDV or you need to create those files from scratch!

Starting with HYDRA MW4.0pe every instance will have an own Oracle Listener configuration.

Logon to the (database) server as local user “oracle”.

Copy the following files from directory /u1/mip1/db_sql/oracle19/network/ to \$ORACLE_HOME/network/admin (e.g. /u1/oracle/ora19/network/admin):

listener.ora
sqlnet.ora
tnsnames.ora

```
cp /u1/mip1/db_sql/oracle19/linux/network/* /u1/oracle/ora19/network/admin
```

Check the configuration file for the Oracle Listener and change contents where necessary, e.g. hostname for the server:

\$ORACLE_HOME/network/admin/listener.ora

```
# listener.ora Network Configuration File: /u1/oracle/ora19/network/admin/listener.ora
# Generated by Oracle configuration tools.

LISTENER_MIP1 =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = TCP) (HOST = 'SERVERNAME') (PORT = 1521))
      (ADDRESS = (PROTOCOL = IPC) (KEY = EXTPROC1521))
    )
  )
```

Attention: For (“HOST = ‘SERVERNAME’”) use the hostname instead of the IP address.

Check the network configuration file “**\$ORACLE_HOME/network/admin/tnsnames.ora**” and change its contents where necessary:

```
MIPl =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP) (HOST = 'SERVERNAME') (PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = mip1)
    )
  )

LISTENER MIPl =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP) (HOST = 'SERVERNAME') (PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = mip1)
      (SID = MIPl)
    )
  )
```

Check the network configuration file “**\$ORACLE_HOME/network/admin/sqlnet.ora**”

Change the access permissions to the files like follows:

```
chmod 644 listener.ora
chmod 644 sqlnet.ora
chmod 644 tnsnames.ora
```

For additional instances (e.g. MIP2, MIP3 etc.) and additional databases (e.g. mip2, mip3 etc.) repeat the above mentioned steps.

6 MIP Database

Every MIP system needs its own MIP database.

Every MIP database needs its own database instance.

Name your MIP databases “mip1”, “mip2”, “mip3”, etc.

Name your MIP database instances “MIP1”, “MIP2”, “MIP3”, etc.

Do not proceed with this chapter if the database installation was meant as preliminary installation for MPDV personnel to install a MIP based product on your server.

Before you might proceed with creating the MIP database(s) on your own you need to have four files available for each MIP database and database instance you want to create.

Example files for MIP database and database instance 1:

CreateMIP1_ora19_lin.sql

MIP1_ora19_lin.sh

MIP1_ora19_lin.sql

initMIP1.ora

Either those files were provided to you in advance by MPDV or you might have the HYDRA application installed at least as far as chapter “4.5 HYDRA Database” according to the HYDRA installation manual:

InstallationGuide_MW40_SLES12.pdf

Note:

The installation manual for HYDRA MW4.0pe will be provided to selected customers only.

6.1 Create MIP Database

Attention:

Do not run these installation procedures on a computer where a MIP based product is already running. By doing so, you might destroy your existing MIP based system. Please be extremely cautious if you decide to do it anyway.

Logon to the server where the database is located as local user "oracle".

Copy the following four files into the directory \$ORACLE_HOME/dbs, e.g. /u1/oracle/ora19/dbs.

```
CreateMIP1_oracle19_lin.sql  
MIP1_oracle19_lin.sh  
MIP1_oracle19_lin.sql  
initMIP1.ora
```

If those files were provided to you by MPDV in advance make sure to copy them to the destination directory, e.g. /u1/oracle/ora19/dbs.

If the HYDRA application might already be installed at least as far as chapter "4.5 HYDRA Database" according to the HYDRA installation manual you can copy those files from the following directory: \$HYDRADIR/db_sql/oracle19/mip1, e.g.: /u1/mip1/db_sql/oracle19/linux/mip1

```
cp /u1/mip1/db_sql/oracle19/linux/mip1/* /u1/oracle/ora19/dbs/
```

Edit all four database creation scripts and make sure that all path configurations are set corresponding to your system environment. All paths used in the database creation scripts must exist on your server! A few of them will be automatically created by the script MIP1_oracle19_lin.sh.

initMIP1.ora

By default, automatic memory management is disabled when you perform typical installation on a node that has more than 4 GB of RAM.

Design fundamentals:

Application server and database sharing the same resources = 30% of available System RAM
Application server and database server on different hardware = 80% of available System RAM

Set the following parameter:

"sga_target" = SGA size desired for the database (70% of the calculated RAM Size)
"pga_aggregate_target" = PGA size desired for the database (30% of the calculated RAM Size)

This are global recommendations of MPDV. For more information about memory settings for Oracle database visit the Oracle documentation.

Depending on the total amount of RAM memory available on the server and the system requirements by the MIP based product please set the size for “sga_target” so that as much memory as possible will be used by the database instance

The size of “sga_target” will have direct influence on the performance of the MIP based application because with bigger sizes more data can be cached by the database.

As long as it is ensured that there will be enough RAM left for the requirements of the server’s operating system (~2GB), the MIP based application services (depending on the system size ~1,5-8GB per instance), the MIP WSP service (depending on the system size ~2-12GB per instance) and other database instances installed on the same server you should use as much memory for “sga_target” as possible.

Make sure that the available server memory (RAM) is used to full capacity without forcing your server into swapping.

Do not change or remove settings regarding user configurations and authorizations or any other permissions without consulting with MPDV first!

By doing so it might be possible that the HYDRA application will not work properly.

Note about database user name “mipadm”:

MPDV strongly recommends not to change MIPs default database user name!

Changing the default database user name is not supported when using Oracle databases!

With Oracle databases it is **not** possible to use different database user names other than MIPs default (CREATE USER MIPADM).

Note about password for database user “mipadm”:

For security reasons MPDV strongly recommends to use a secure password for the database user mipadm!

The new password must be disclosed to MPDV.

The following characters are **not** allowed for the password:

- Characters with ASCII Code > 126 (e.g. German umlauts, French “umlauts”, etc.)
- Pipe: |
- Ampersand: &

To install additional database instances, e.g. MIP2, there are corresponding scripts available, e.g.:

```
CreateMIP2_ora19_lin.sql  
MIP2_ora19_lin.sh  
MIP2_ora19_lin.sql  
initMIP2.ora
```

Example files can be found in a HYDRA installation, e.g. in: /u1/mip1/db_sql/oracle19/linux/mip2

When installing additional database instances make sure to change the port settings for “EM Express” in file “createMIP_x_ora19_lin.sql” so that there will be no port conflicts, e.g.:

```
MIP1:      exec dbms_xdb_config.sethttpsport(5500) ;  
MIP2:      exec dbms_xdb_config.sethttpsport(5501) ;
```

Change to directory \$ORACLE_HOME/dbs, e.g.:

```
cd /u1/oracle/ora19/dbs
```

```
chmod u+x CreateMIP1_oral9_lin.sh
```

```
./CreateMIP1_oral9_lin.sh
```

Enter ORACLE_BASE (/u1/oracle) Hit 'Return' if you want to use the default values :

Enter ORACLE_HOME: (/u1/oracle/ora19) Hit 'Return' if you want to use the default values :

Enter database service name (mip1) Hit 'Return' if you want to use the default values :

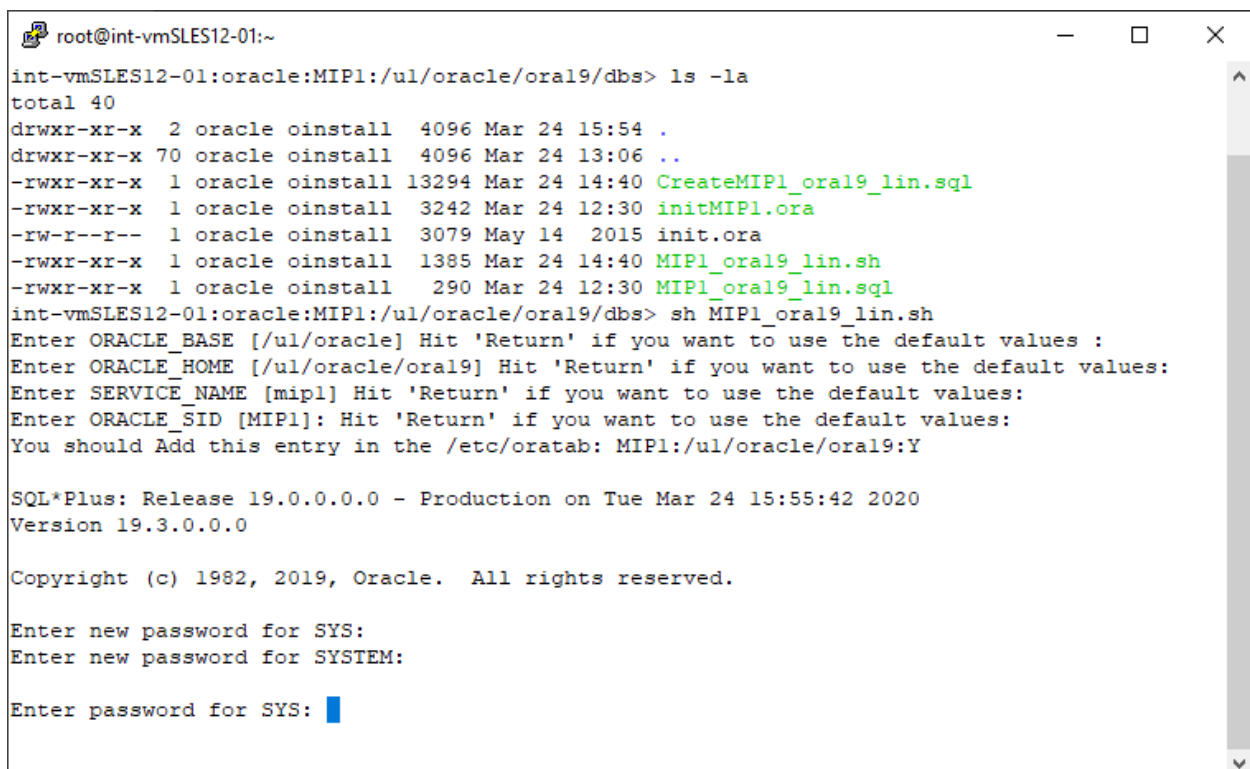
Enter ORACLE_SID (MIP1) Hit 'Return' if you want to use the default values :

Enter new password for SYS: MW40pe!!2020

Enter new password for SYSTEM: MW40pe!!2020

Enter password for SYS: MW40pe!!2020

Example for database instance „MIP1“ and database „mip1“:



```
root@int-vmSLES12-01:~
int-vmSLES12-01:oracle:MIP1:/u1/oracle/ora19/dbs> ls -la
total 40
drwxr-xr-x  2 oracle oinstall  4096 Mar 24 15:54 .
drwxr-xr-x 70 oracle oinstall  4096 Mar 24 13:06 ..
-rwxr-xr-x  1 oracle oinstall 13294 Mar 24 14:40 CreateMIP1_oral9_lin.sql
-rwxr-xr-x  1 oracle oinstall  3242 Mar 24 12:30 initMIP1.ora
-rw-r--r--  1 oracle oinstall  3079 May 14 2015 init.ora
-rwxr-xr-x  1 oracle oinstall  1385 Mar 24 14:40 MIP1_oral9_lin.sh
-rwxr-xr-x  1 oracle oinstall   290 Mar 24 12:30 MIP1_oral9_lin.sql
int-vmSLES12-01:oracle:MIP1:/u1/oracle/ora19/dbs> sh MIP1_oral9_lin.sh
Enter ORACLE_BASE [/u1/oracle] Hit 'Return' if you want to use the default values :
Enter ORACLE_HOME [/u1/oracle/ora19] Hit 'Return' if you want to use the default values:
Enter SERVICE_NAME [mip1] Hit 'Return' if you want to use the default values:
Enter ORACLE_SID [MIP1]: Hit 'Return' if you want to use the default values:
You should Add this entry in the /etc/oratab: MIP1:/u1/oracle/ora19:Y

SQL*Plus: Release 19.0.0.0.0 - Production on Tue Mar 24 15:55:42 2020
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Enter new password for SYS:
Enter new password for SYSTEM:

Enter password for SYS: █
```

At the end of the installation process you must add the entry in the `/etc/oratab` for an automatic start of the database after starting or rebooting of the database server.

Depending on your server environment the installation will run for at least 10 - 20 minutes or longer.

Under no circumstances you should interrupt the installation process!

The installation is finished as soon as „create EM Express“ is through and the Linux prompt is visible again, e.g.:



```
root@int-vmSLES12-01:~  
Total System Global Area 3237999728 bytes  
Fixed Size                8901744 bytes  
Variable Size             654311424 bytes  
Database Buffers          2566914048 bytes  
Redo Buffers              7872512 bytes  
Database mounted.  
Database opened.  
SQL> spool off  
SQL>  
SQL> -- create EM Express  
SQL> -- -----  
SQL> SET VERIFY OFF  
SQL> connect "SYS"/"&&sysPassword" as SYSDBA  
Connected.  
SQL> spool /u1/oracle/admin/mip1/scripts/13-em_express.log  
SQL> exec dbms_xdb_config.sethttpsport(5501);  
  
PL/SQL procedure successfully completed.  
  
SQL> spool off  
SQL>  
SQL> exit;  
Disconnected from Oracle Database 19c Standard Edition 2 Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0  
int-vmSLES12-01:oracle:MIP1:/u1/oracle/ora19/dbs>
```

The database creation must complete without errors.

Check all log files of the installation which can be found in the following directory:

\$ORACLE_BASE/admin/<database name>/scripts

e.g. `/u1/oracle/admin/mip1/scripts/`

Start the Oracle Listener:

```
lsnrctl start LISTENER_MIP1
```

```

MOS-PD-vLIN-21
MOS-PD-vLIN-21:oracle:MIPI:/ul/oracle> lsnrctl start LISTENER_MIPI

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 09-MAR-2020 11:13:31

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Starting /ul/oracle/oral9/bin/tnslsnr: please wait...

TNSLSNR for Linux: Version 19.0.0.0.0 - Production
System parameter file is /ul/oracle/oral9/network/admin/listener.ora
Log messages written to /ul/oracle/diag/tnslsnr/MOS-PD-vLIN-21/listener_miPI/alert/log.xml
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp) (HOST=MOS-PD-vLIN-21.mpdv.local) (PORT=1521)))
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc) (KEY=EXTPROC1521)))

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP) (HOST=MOS-PD-vLIN-21.mpdv.local) (PORT=1521)))
STATUS of the LISTENER
-----
Alias                     LISTENER_MIPI
Version                   TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date                09-MAR-2020 11:13:31
Uptime                   0 days 0 hr. 0 min. 0 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   /ul/oracle/oral9/network/admin/listener.ora
Listener Log File         /ul/oracle/diag/tnslsnr/MOS-PD-vLIN-21/listener_miPI/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp) (HOST=MOS-PD-vLIN-21.mpdv.local) (PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc) (KEY=EXTPROC1521)))
The listener supports no services
The command completed successfully
MOS-PD-vLIN-21:oracle:MIPI:/ul/oracle>

```

Check the status after a few minutes

```

MOS-PD-vLIN-21
MOS-PD-vLIN-21:oracle:MIPI:/ul/oracle> lsnrctl status LISTENER_MIPI

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 09-MAR-2020 11:15:33

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP) (HOST=MOS-PD-vLIN-21.mpdv.local) (PORT=1521)))
STATUS of the LISTENER
-----
Alias                     LISTENER_MIPI
Version                   TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date                09-MAR-2020 11:13:31
Uptime                   0 days 0 hr. 2 min. 1 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   /ul/oracle/oral9/network/admin/listener.ora
Listener Log File         /ul/oracle/diag/tnslsnr/MOS-PD-vLIN-21/listener_miPI/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp) (HOST=MOS-PD-vLIN-21.mpdv.local) (PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc) (KEY=EXTPROC1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp) (HOST=MOS-PD-vLIN-21.mpdv.local) (PORT=5500) (Security=(my_wallet_directory=/ul/oracle/admin/
miPI/xdw_wallet)) (Presentation=HTTP) (Session=RAW)))
Services Summary...
Service "MIPIXDS" has 1 instance(s).
  Instance "MIPI", status READY, has 1 handler(s) for this service...
Service "miPI" has 1 instance(s).
  Instance "MIPI", status READY, has 1 handler(s) for this service...
The command completed successfully
MOS-PD-vLIN-21:oracle:MIPI:/ul/oracle>

```

Use **sqlplus** to test the connection to the database(s).

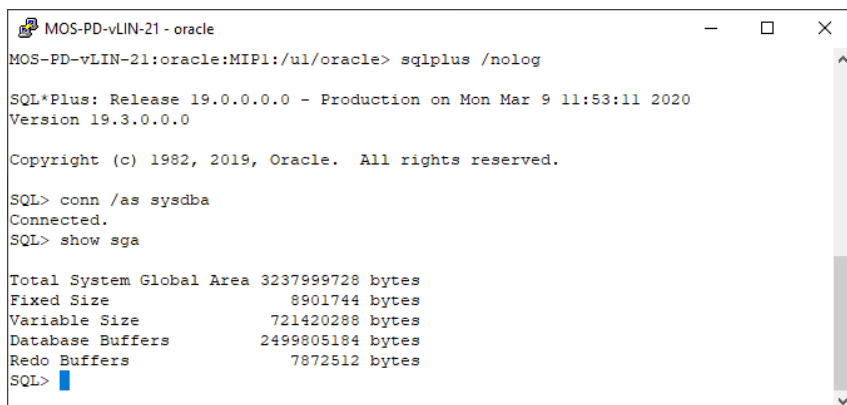
```

sqlplus /nolog

connect / as sysdba

SQL> show sga

```



```
MOS-PD-vLIN-21 - oracle
MOS-PD-vLIN-21:oracle:MIP1:/u1/oracle> sqlplus /nolog

SQL*Plus: Release 19.0.0.0.0 - Production on Mon Mar 9 11:53:11 2020
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

SQL> conn /as sysdba
Connected.
SQL> show sga

Total System Global Area 3237999728 bytes
Fixed Size                  8901744 bytes
Variable Size              721420288 bytes
Database Buffers          2499805184 bytes
Redo Buffers                7872512 bytes
SQL>
```

SQL> **exit**

Logon to the server as local user “mipadm”.

Make a backup copy of all the files you used to create your HYDRA database, e.g.:

```
cp /u1/oracle/ora19/dbs/createMIP1_ora19_lin.sql /u1/mip1/db_sql
cp /u1/oracle/ora19/dbs/MIP1_ora19_lin.sh /u1/mip1/db_sql
cp /u1/oracle/ora19/dbs/MIP1_ora19_lin.sql /u1/mip1/db_sql
cp /u1/oracle/ora19/dbs/initMIP1.ora /u1/mip1/db_sql
```

Repeat these procedures for every HYDRA database and its instance.

Should there have been a problem while creating the database you might need to run the whole process again after you fixed the problems which occurred the first time.

It might be that the instance is already running after a faulty attempt.

Before you can try it a second time you need to shut down the instance first and you need to delete already created control files.

Shut down the database:

```
sqlplus /nolog
connect / as sysdba
SQL> shutdown abort
SQL> exit
```

Delete already created control files, e.g.:

```
rm /u1/oracle/oradata/mip1/MIP1_control01.ctl
rm /u1/oracle/fast_recovery_area/mip1/MIP1_control02.ctl
rm -rf /u1/oracle/oradata/mip1
```

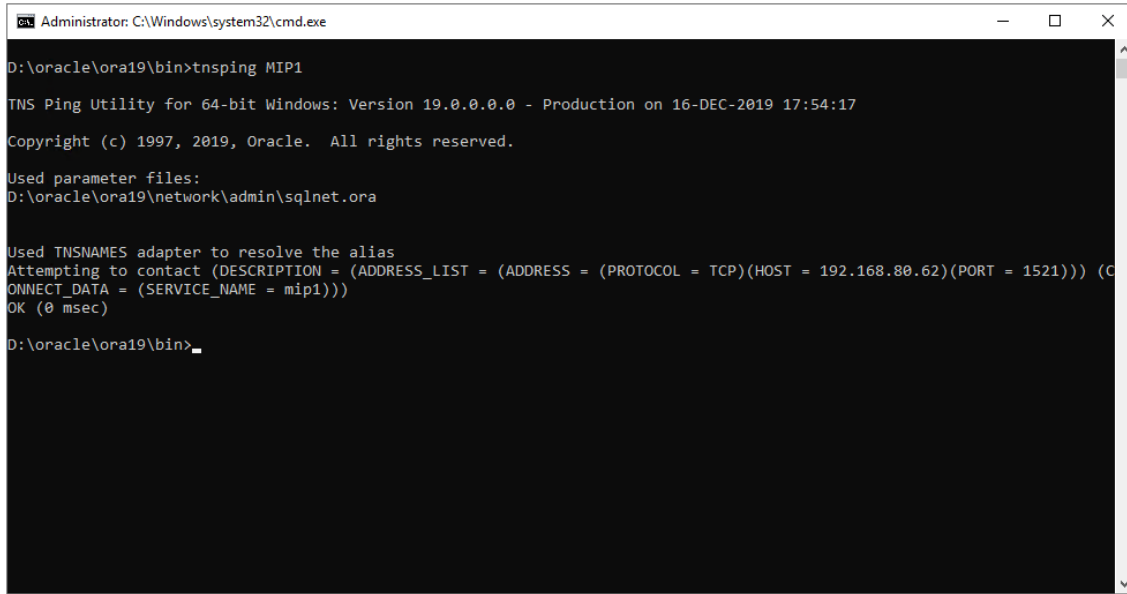
Now you might run the database creation process again.

6.2 Check Database Connection

Check that the new database is accessible using the **tnsping** command.

Open a command prompt:

```
/u1/oracle/ora19/bin/tnsping mip1
```



```
Administrator: C:\Windows\system32\cmd.exe
D:\oracle\ora19\bin>tnsping MIP1

TNS Ping Utility for 64-bit Windows: Version 19.0.0.0.0 - Production on 16-DEC-2019 17:54:17

Copyright (c) 1997, 2019, Oracle. All rights reserved.

Used parameter files:
D:\oracle\ora19\network\admin\sqlnet.ora

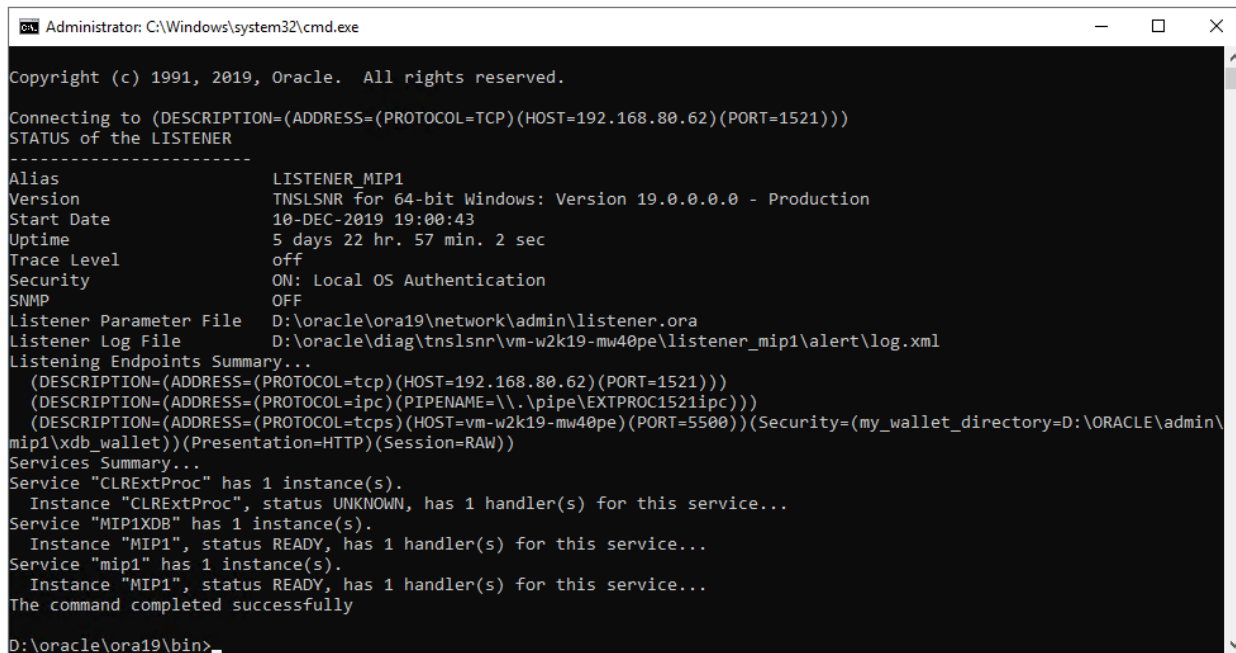
Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS_LIST = (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.80.62)(PORT = 1521))) (C
ONNECT_DATA = (SERVICE_NAME = mip1)))
OK (0 msec)

D:\oracle\ora19\bin>_
```

6.3 Check Oracle Listener

After installing the HYDRA database(s) MIP1 (MIP2, MIP3, etc.) check that all services are running using the **lsnrctl** command.

```
/u1/oracle/ora19/bin/lsnrctl status LISTENER_MIP1
```



```
Administrator: C:\Windows\system32\cmd.exe

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.80.62)(PORT=1521)))
STATUS of the LISTENER
-----
Alias                     LISTENER_MIP1
Version                  TNSLSNR for 64-bit Windows: Version 19.0.0.0 - Production
Start Date               10-DEC-2019 19:00:43
Uptime                   5 days 22 hr. 57 min. 2 sec
Trace Level              off
Security                 ON: Local OS Authentication
SNMP                     OFF
Listener Parameter File  D:\oracle\ora19\network\admin\listener.ora
Listener Log File        D:\oracle\diag\tnslsnr\vm-w2k19-mw40pe\listener_mip1>alert\log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=192.168.80.62)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(PIPENAME=\\.\pipe\EXTPROC1521ipc)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcps)(HOST=vm-w2k19-mw40pe)(PORT=5500))(Security=(my_wallet_directory=D:\ORACLE\admin\mip1\xdb_wallet))(Presentation=HTTP)(Session=RAW))
Services Summary...
Service "CLRExtProc" has 1 instance(s).
  Instance "CLRExtProc", status UNKNOWN, has 1 handler(s) for this service...
Service "MIP1XDB" has 1 instance(s).
  Instance "MIP1", status READY, has 1 handler(s) for this service...
Service "mip1" has 1 instance(s).
  Instance "MIP1", status READY, has 1 handler(s) for this service...
The command completed successfully

D:\oracle\ora19\bin>
```

Check if a service “mip1” is available

Additional listener could be called by its service name (LISTENER_MIP2, LISTENER_MIP2)

6.4 Check Automatic Startup Database

Logon to the HYDRA (database) server as local user “root”.

For an automatic startup of the Oracle database, the Listener and the Enterprise Manager the following entries must be set to “yes” in the file `/etc/sysconfig/oracle`:

```
START_ORACLE_DB="yes"  
START_ORACLE_DB_LISTENER="yes"  
START_ORACLE_DB_EMANAGER="yes"
```

Those settings should already be done if your server was properly configured according to the following MPDV manual:

"Preparation Guide for SLES12 SP5" (e.g.: `PreInstGuide_MW40_SLES12_SP5.pdf`)

Additionally in file `/etc/oratab` there must be an entry for every Oracle instance (e.g. MIP1 and MIP2) with the correct path for \$ORACLE_HOME and a “Y” at the end of each line, e.g.:

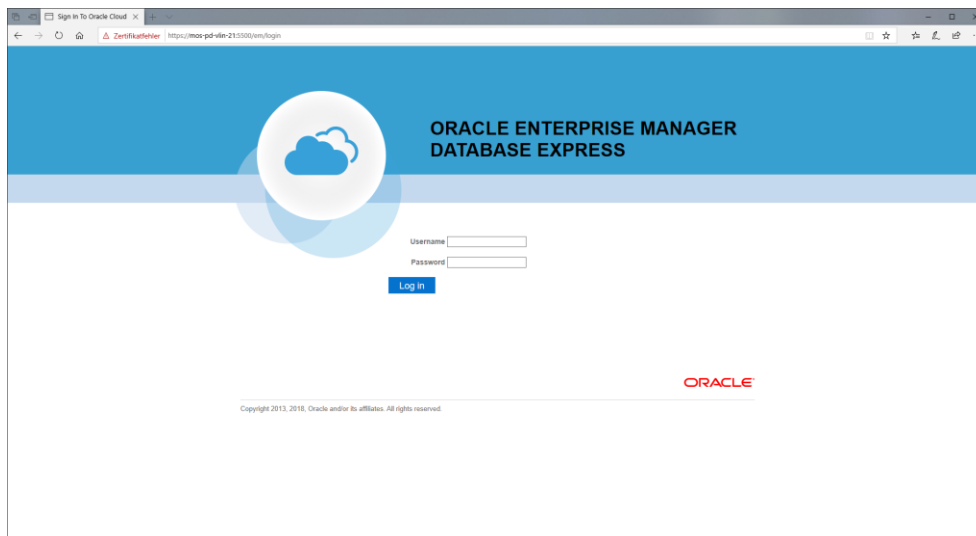
```
MIP1:/u1/oracle/ora19:Y  
MIP2:/u1/oracle/ora19:Y
```

6.5 Check Oracle Enterprise Manager

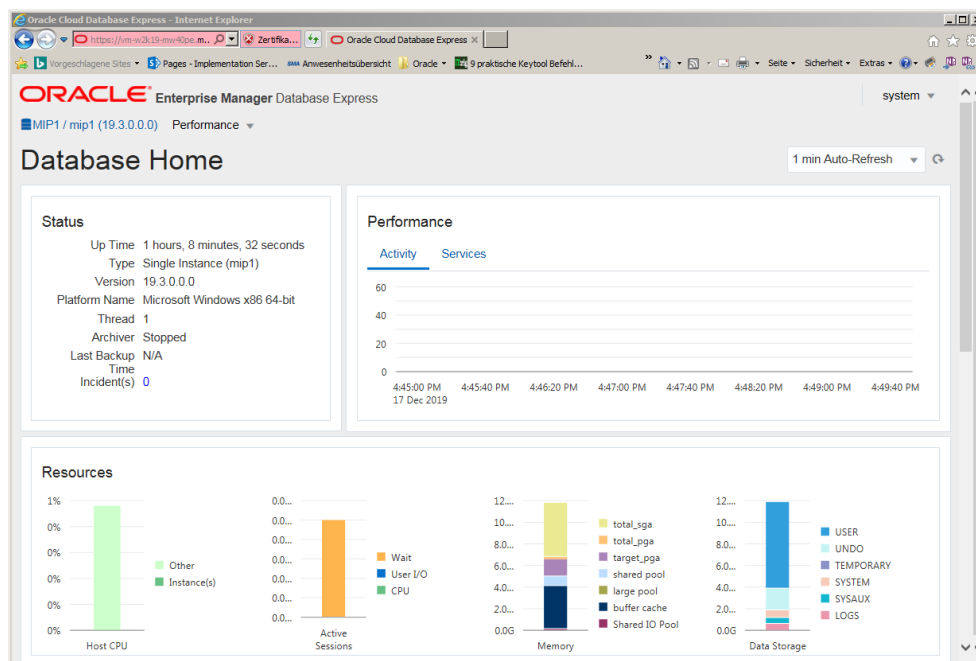
After a successful installation you can test the connection to the Oracle Enterprise Manager (Database Express) by using the Internet Explorer (e.g. to <https://servername:5500/em>).

Note: For using the Oracle Enterprise Manager (Database Express) a web browser with Adobe Flash Player Plugin is needed.

The Oracle Enterprise Manager (Database Express) is only available if the database is running.



Login with user “system” and the defined password

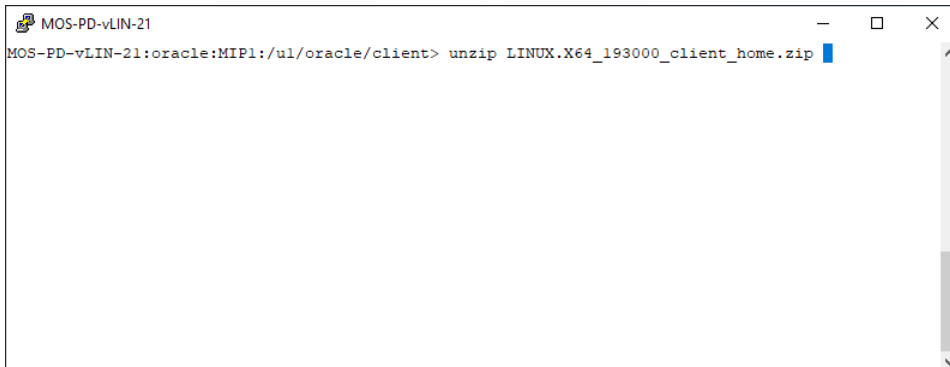


7 Oracle 64Bit Client on Application Server

With **separate application and database servers** for MIP based products the Oracle Client software must **always** be installed on the **application server**.

Login as user **oracle**

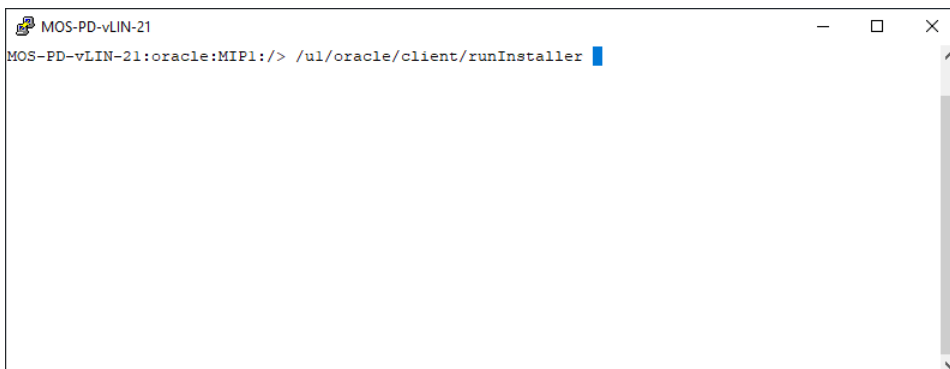
Unzip the Oracle client installation file to the destination directory (e.g.: /u1/oracle/client)

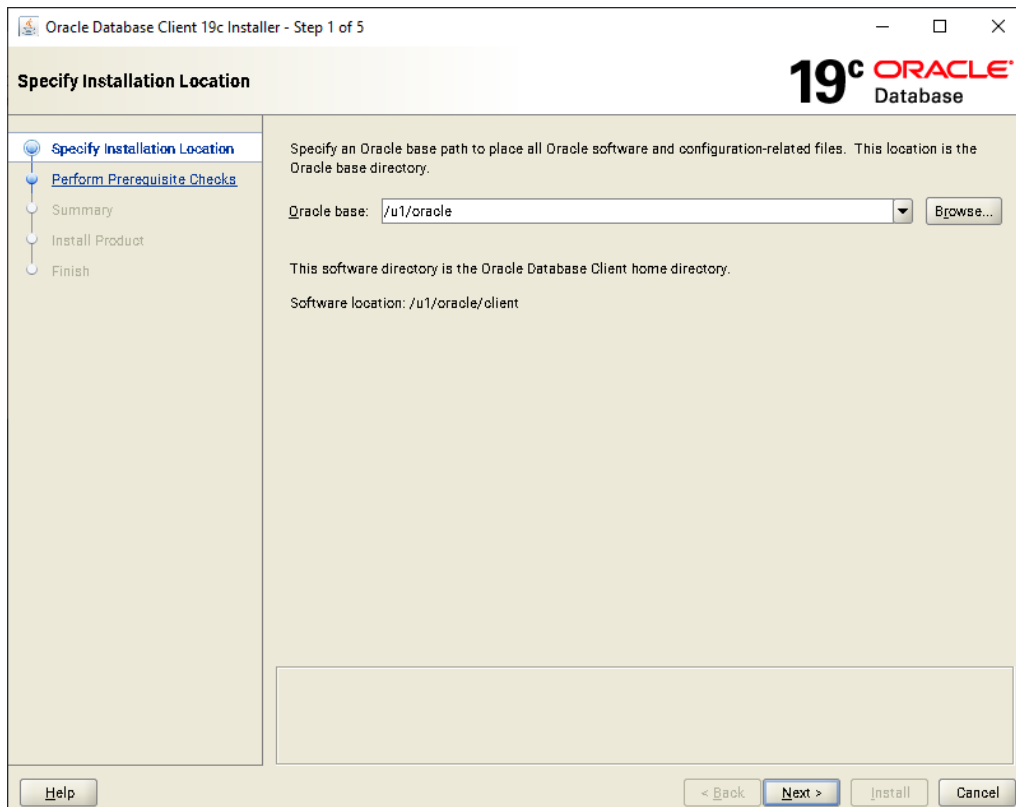
A terminal window titled 'MOS-PD-vLIN-21' with standard window controls. The command prompt shows the user 'oracle' at host 'MIP1' in directory '/u1/oracle/client'. The command 'unzip LINUX.X64_193000_client_home.zip' has been entered, and a blue cursor is at the end of the line.

```
MOS-PD-vLIN-21:oracle:MIP1:/u1/oracle/client> unzip LINUX.X64_193000_client_home.zip
```

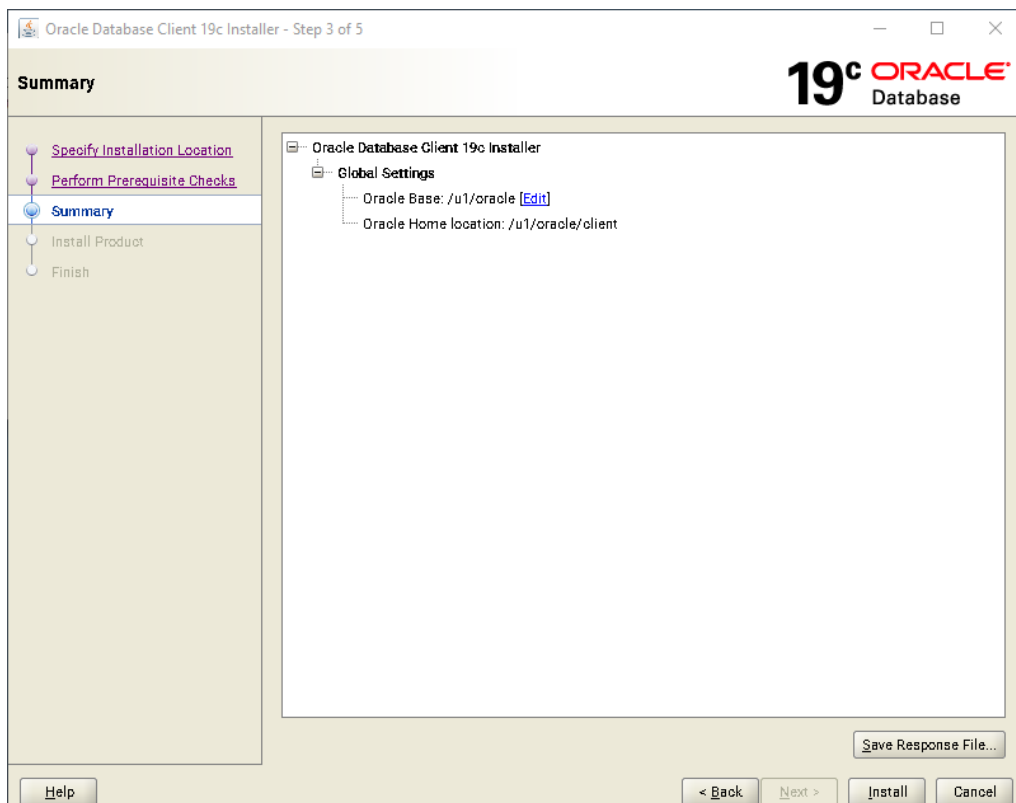
Start the installation

/u1/oracle/client/runInstaller

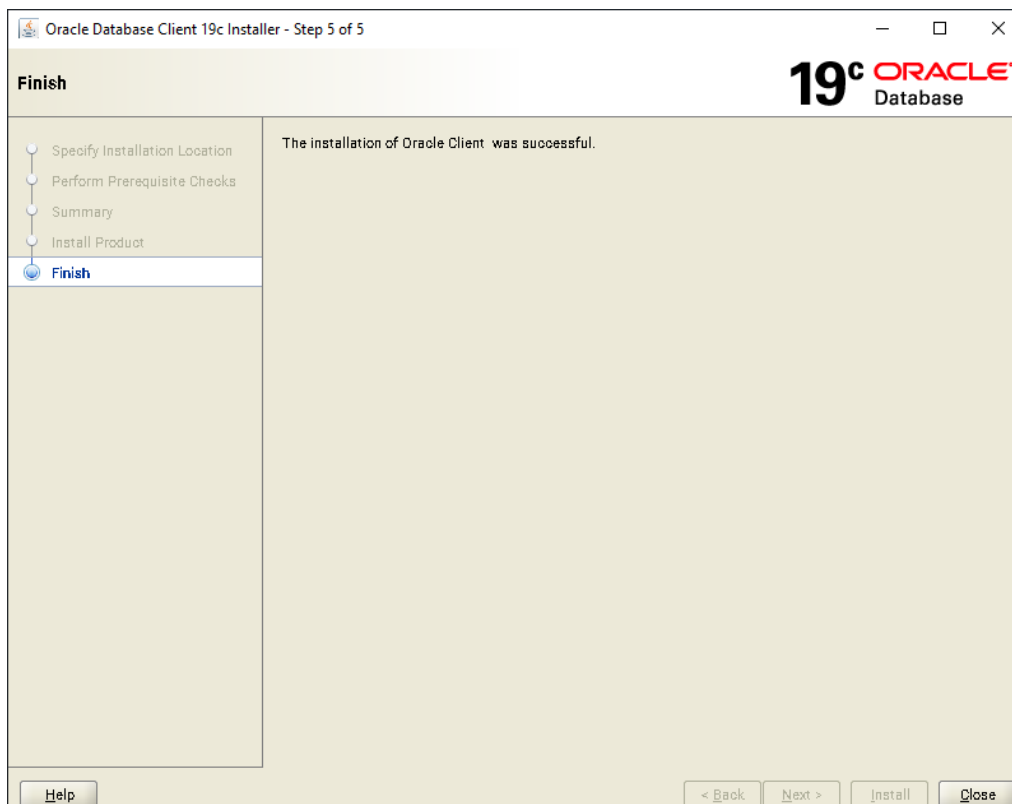
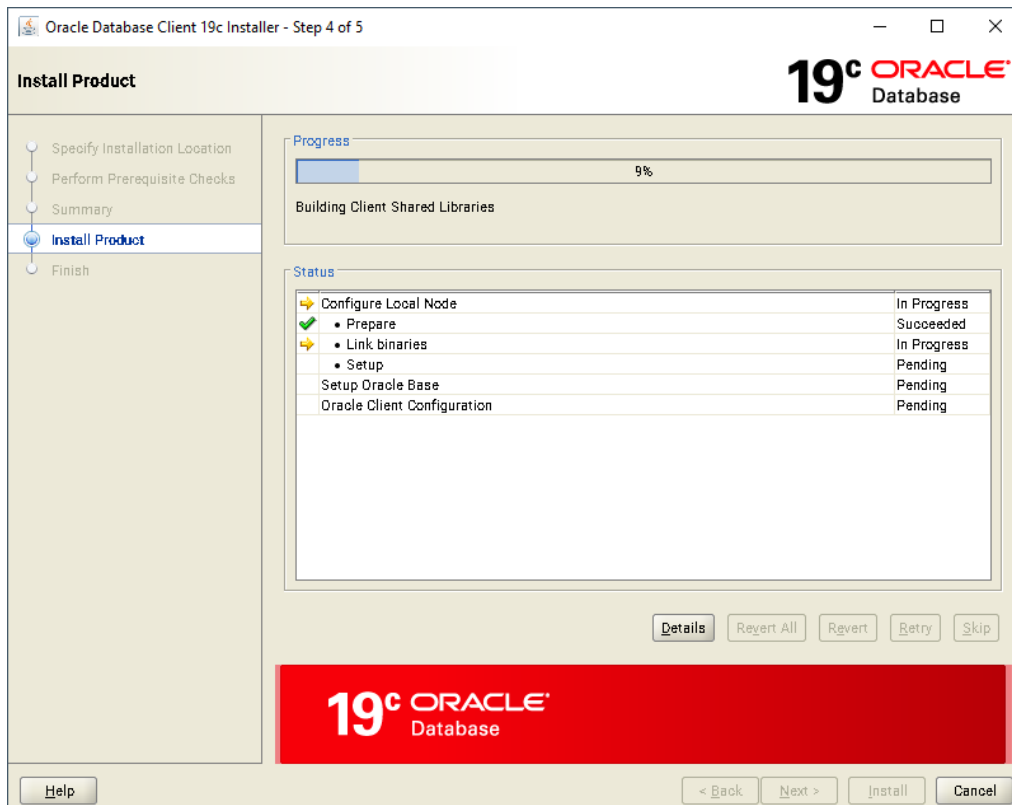
A terminal window titled 'MOS-PD-vLIN-21' with standard window controls. The command prompt shows the user 'oracle' at host 'MIP1' in directory '/> /u1/oracle/client/runInstaller



Next



Install



Close

8 Post installation configuration

8.1 tnsnames.ora

Copy the content of /u1/mip1/db_sql/oracle19/linux/network/client/* to /u1/oracle/client/network/admin*

On the HYDRA application server check the **tnsnames.ora** file of the client and add necessary alias configurations for the HYDRA databases (Default: MIP1):

/u1/oracle/client/network/admin/tnsnames.ora

```
# tnsnames.ora Network Configuration File: /u1/oracle/client/network/admin/tnsnames.ora
# Generated by Oracle configuration tools.
```

```
MIP1 =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP) (HOST = [IP ADDRESS or FQDN NAME]) (PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = mip1)
    )
  )

MIP2 =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP) (HOST = [IP ADDRESS or FQDN NAME]) (PORT = 1522))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = mip2)
    )
  )
```

9 Additional Documents

Oracle Database Installation Guide, 19c for Microsoft Windows:

Database Installation Guide for Linux

<https://docs.oracle.com/en/database/oracle/oracle-database/19/ladbi/index.html>

Database Client Installation Guide for Linux

<https://docs.oracle.com/en/database/oracle/oracle-database/19/lacli/index.html>

Database Administrator's Guide

<https://docs.oracle.com/en/database/oracle/oracle-database/19/admin/index.html>